3RD SEM./ MINING /2025(W)NEW

TH3 Mine Geology-I

Full Marks: 70 **Time- 3 Hrs**

Answer any four Questions including Q No.1& 2

Figures in the right hand margin indicates marks

1. Answer **All** questions 2 x 10

a. Define weathering and denudation.

b. Define Amygdaloidal and vesicular Structure with example.

c. Differentiate Colour & Lustre with example.

d. Define metamorphism & its agents, write down two examples of metamorphic rock.

e. Define ore forming mineral & name two oxide mineral.

f. Mantle & Core are extended up to what depth & name the discontinuity present between MANTLE & CORE.

g. What is interfacial Angle? Name the Instrument used to Measure the Interfacial Angle.

h. Construct figure on True dip and Strike of a bed and define them.

i. If Dip of a Fault plane is 45^0, find out the Hade of that Fault Plane.

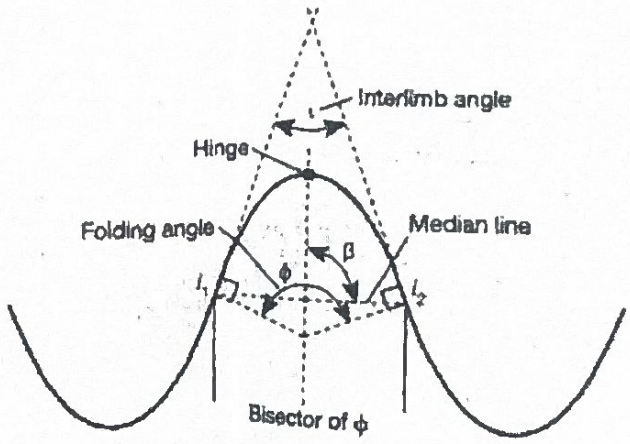
j. Name any two erosional landforms produce due to geological action of wind and explain them.

2. Answer **Any Six** Questions 5X6

a. What is dune? Describe the different types of dune with diagram.

b. Describe Mho's hardness scale and write down the 10 minerals present in the scale. Name the mineral which shows double hardness.

c. i) Define fold. What is the interlimb angle of an isoclinal fold?
ii)If interlimb angle of a fold is 60^0, Find the folding angle.



d. What is crystallography? Briefly describe the different crystal system on the basis of crystallographic axes.

e. What is texture? Describe with neat sketches (if needed) the various textures on the basis of crystallinity, Grain size in igneous rocks.

f. What is Unconformity? Describe the difference stages of formation of Angular unconformity & Disconformity with diagram.

g A rock is made up of 100% pure SiO_2. Calculate the mass % of Si in the Rock. Given Atomic Mass of Si= 28 g/mol, Atomic Mass of O= 16 g/mol

Answer **Any TWO** Questions

3. What is Rock? Classify different rock Groups with example. 10

4. Define Mineral. Describe the various physical properties of minerals with example. 10

5. Discuss Briefly the Geological work done by a Wind & Describe in brief the different landforms produced due to Wind action. 10

6. Classify & describe with neat sketches (if needed) the different types of Faults. 10